J. Dairy Sci. 109:6416–6425
https://doi.org/10.3168/jds.2025-27459

Genetic analysis of resilience in Israeli Holstein cattle based on first parity daily rumination and milk production records

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ABSTRACT

Resilience in farm animals was proposed as an indicator trait for health and robustness. One approach is to analyze the variance and autocorrelation of daily milk production. Daily rumination time is a useful indicator of health status. Recently developed rumination monitoring tags provide accurate daily output data on large numbers of cows. Genetic correlations were computed between the means, log variances, and autocorrelations of daily milk production and rumination time, 5 metabolic diseases (retained placenta, metritis, ketosis, displaced abomasum, and milk fever), 7 traits included in the Israeli selection index, and BCS near calving. The first 40 DIM and the remainder of the lactation of 40,860 first-parity cows calving between 2019 and 2024 were analyzed separately. Significant genetic correlations of the resilience indicator traits were found for all 5 diseases, BCS, and all 7 index traits. The log variance of daily milk production during the first 40 DIM had a heritability of 0.11 and was the best resilience indicator among those tested. A correlation of −0.62 was found between this trait and ketosis incidence. Coefficients of determination for the breeding values for the Israeli breeding index based on 7 indicator traits computed from 1 to 40 DIM were 0.18 for cows and 0.19 for sires. Inclusion of a resilience predictor trait based on the first 40 DIM in the selection index is justified to improve selection decisions for young cows and sires.
Key words: dairy cattle, genetic analysis, resilience, rumination time, body condition score

INTRODUCTION

Resilience was proposed as an indicator trait for health and robustness in farm animals, and several resilience indicators in dairy cattle have been proposed. Body condition score reflects the cow's energy balance and is related to disease resistance and tolerance (Calus and Veerkamp, 2003; König and May, 2019; Bengtsson et al., 2022), and this trait was recently analyzed in detail by Weller and Ezra (2025) on Israeli Holsteins. Bengtsson et al. (2022) proposed using BCS as an indicator trait for resilience. Another approach for estimating resilience is to analyze the variance and autocorrelation of a longitudinal recorded trait. For dairy cattle, this has generally been daily milk production. The underlying assumption is that an animal responds to a disturbance with a change in performance: "As it is generally not known when a disturbance begins, when it ends, and whether it exists at all, it is not the direct response to a specific disturbance which is analyzed, but the change of the longitudinal trait over time. The variance of daily milk production over time should therefore quantify resilience, with a low variance indicative of high resilience" (Keßler et al., 2025, p. 726). It has been proposed that the autocorrelation indicates how quickly an individual recovers from a disturbance, with an autocorrelation close to zero suggesting a high level of resilience (Berghof et al., 2019; Poppe et al., 2021; Friggens et al., 2022). Meijer et al. (2024) proposed including resilience indicator traits in routine breeding value estimation.

No gold standard has been established for which resilience indicator traits should be included in the total merit index. This is because the underlying disturbance is usually not known, making the derivation of economic weights difficult. One approach is to analyze genetic correlations of resilience indicator traits with health and production traits, length of productive life, or overall performance, which can be used to estimate their economic values. Some previous studies have done this by estimating the correlation of resilience indicator traits with existing selection indices (Elgersma et al., 2018; Poppe et al., 2021) or their correlations with disease prevalence (e.g., reproductive disorders; Wang et al., 2022) in dairy cattle. Those studies showed that more resilient animals tend to be healthier. The different sensitivity of various resilience indicator traits is indicated by the example of mastitis. Although the variance of daily milk yield tended to be higher in diseased individuals, as expected, the au-

Received August 20, 2025.
Accepted February 20, 2026.
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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Table 1. Numbers of first-parity records, cows, and means of daily rumination time and milk production by calving year

Calving year	Records	Cows	Rumination time (min)	Milk yield (kg)
2019	1,047,639	7,547	526.68	33.59
2020	3,211,826	11,868	523.88	33.41
2021	3,231,379	11,879	525.85	33.91
2022	3,416,360	13,046	524.70	33.52
2023	3,638,436	13,894	519.46	34.09
2024	3,501,865	15,580	527.79	34.46
Total/mean[1]	18,047,505	73,814	524.42	33.87

[1]Totals for records and cows and means for rumination time and milk yield.

tocorrelation was lower (Kok et al., 2021). Keßler et al. (2025) proposed a resilience selection index composed of 2 different variance-based resilience indicator traits. Its correlation with overall performance was positive, but all correlations were <0.5. Incorporating more than 2 resilience indicator traits into the index improved the correlated response in health traits only slightly.

Daily rumination time (**RT**) has been proposed as an indicator of the health status in dairy cattle. Rumination time is associated with the metabolic condition and disease state of dairy cattle near parturition. Therefore, RT monitoring may be helpful to quickly obtain information on the health status of animals in this critical period (Mammi et al., 2021; Siivonen et al., 2011; Soriani et al., 2012). Recently developed rumination monitoring tags (SCR Engineers Ltd., Netanya, Israel) provide accurate daily output data on large numbers of cows. Weller and Ezra (2024) estimated environmental factors affecting RT in high yielding dairy cattle as recorded by SCR monitoring tags, genetic parameters for RT across parities, and environmental and genetic correlations between RT and economic traits on a sample of more than 30 million daily records from over 77,000 Israeli Holstein cows. They also predicted the consequence of inclusion of this trait in the Israeli breeding index. All changes in expected gains due to inclusion of RT in the index were economically positive, except for fat and SCS. Inclusion of RT in the index should result in 1 kg less gain in fat, a miniscule gain of 0.03 for SCS; and gains of 1.5 kg protein, 0.3% female fertility, and 5-d herd life.

Both mean milk production and RT vary during the course of the lactation. In first-parity cows both traits increase rapidly during the first 2 mo in milk, reach a peak during the second or third month, and then decline slowly during the remainder of the lactation. Poppe et al. (2021) attempted to estimate resilience based on natural log-transformed variance and lag-1 autocorrelation, with each trimester of the total lactation analyzed separately. Because of the distinctive shape of the lactation curves for both traits, we proposed analyzing the first 40 DIM and the remainder of the lactation separately. The objectives of the current study were to estimate genetic and environmental correlations between these 2 resilience indicators, BCS and means of daily milk production and RT, 5 metabolic diseases (retained placenta, metritis, ketosis, displaced abomasum, and milk fever), and 7 traits included in the Israeli breeding index. In addition, we estimated the utility of breeding values of these resilience indicators to predict breeding values of the diseases, the index traits, and the complete Israeli selection index.

MATERIALS AND METHODS

Datasets Analyzed

The basic dataset consisted of 18,047,505 first-parity records of daily milk and rumination records of 73,814 cows, from 168 herds, collected from 2019 through 2024. Daily milk production and RT were recorded, using the SCR recording system (SCR Engineers Ltd., Netanya, Israel). Records from parturition through 305 DIM were retained. Numbers of valid records, cows and mean RT and daily milk production by calving year are given in Table 1. Overall mean daily RT was 524 min and mean daily milk production was 34 kg. Mean RT as a function of DIM is plotted in Figure 1. Daily RT and milk increase rapidly after parturition until ~40 DIM, then continue to increase slowly until a peak at ~75 DIM for RT and 90 DIM for milk.

For the computation of lactation records, the following edits were applied to RT: Cow records with daily RT <50 and RT >1,000 min and daily milk <5 and milk >100 kg were deleted. In the analysis of DIM 1 to 40, only lactations with at least 20 daily records, first record no more than 10 DIM, and last record at least 30 DIM in were retained. Similarly, for the analysis of 41 to 305 DIM, only lactations with at least 100 daily records, first record no later than 50 DIM, and last record at least 200 DIM were retained. Cow records were retained if

1. There were valid records for both the first 40 DIM and the remainder of the lactation.

Table 2. Mean and SD of the phenotypic values for disease and index traits analyzed

Trait type	Trait	Mean	SD	Index coefficient	Fraction of the index[1]
Disease	Retained placenta (%)	2.19	14.65	0	0
	Metritis (%)	45.10	49.76	0	0
	Ketosis (%)	9.47	29.29	0	0
	Displaced abomasum (%)	0.24	4.87	0	0
	Milk fever (%)	0.04	2.10	0	0
	BCS	3.10	0.34	0	0
Index	Milk (kg)	12,627	1,707	0	0
	Fat (kg)	480.73	66.08	9.94	0.25
	Protein (kg)	410.83	51.39	19.88	0.36
	SCS[2]	3.16	1.17	−300	0.12
	Fertility[3] (%)	47.79	34.88	26	0.13
	Herd life (d)	1,138	485	0.6	0.07
	Persistency[4] (%)	68.64	12.59	10	0.04

[1]Computation of the fraction of the index explained by each trait is given in Weller and Ezra (2024, 2025).

[2]Units for SCS are based on the $\log_2$ frequency of somatic cells/mL, as described by Weller and Ezra (1997). Negative values are economically favorable.

[3]The fertility trait is "conception status," as defined in Weller and Ezra (2004).

[4]Persistency as defined by Weller et al. (2006).

2. There were valid records for the 5 metabolic diseases analyzed: retained placenta, metritis, ketosis, displaced abomasum, and milk fever.
3. Body condition score was recorded from 14 d before calving to 14 DIM.
4. Calving age was between 21 and 29 mo.
5. Sire of the cow was listed as Holstein.

After these edits, 40,860 cows were retained for further analysis. Methods used for scoring the disease traits are given in Weller et al. (2019). The 5 disease traits were scored dichotomously; either the disease was recorded by the first 40 DIM (score = 1) or not (score = 0). Body condition was scored over the range 1.50 to 4.75 by increments of 0.25 (Weller and Ezra, 2025).

The 9 traits included in the Israeli breeding index (**PD20**) are milk, fat, and protein production, SCS, female fertility, herd life, milk production persistency, calving ease, and calf survival (Weller and Ezra, 2024, 2025). Genetic correlations were not computed for calving ease and calf survival because calving ease was included as a fixed effect in the analysis models, based on the assumption that a difficult calving is a traumatic experience for the cow and should therefore have a direct effect on resilience. Four levels of calving ease were defined: easy, difficult, unassisted (the calving was not observed), and surgical. First-parity heritabilities for calving ease and calf survival in the Israeli Holstein population were estimated as 0.030 and 0.014 (Weller and Ezra, 2016).

The analysis included the first-parity records for 6 of the 9 traits included in PD20. For herd life, only a single record is generated per cow. Methods used to compute the phenotypic values of the index traits are described by Weller and Ezra (2004, 2016, 2016) and Weller et al. (2006). For female fertility, the trait analyzed was "conception status," as defined by Weller and Ezra (2004). Breeding values for calving ease and calf survival are computed based on first and second calving records (Weller and Ezra, 2016). The means and SD of the phenotypic values for the disease and index traits analyzed and the index coefficients and fraction of the index explained by each trait are given in Table 2.

Body condition score is listed with the disease traits because it was included in the same dataset. Frequencies of displaced abomasum and milk fever were both <1%. Because the sample of animals analyzed included only cows with first calving between 2019 and 2024, most of the results for herd life are based on "expected herd life," as computed by Weller and Ezra (2015).

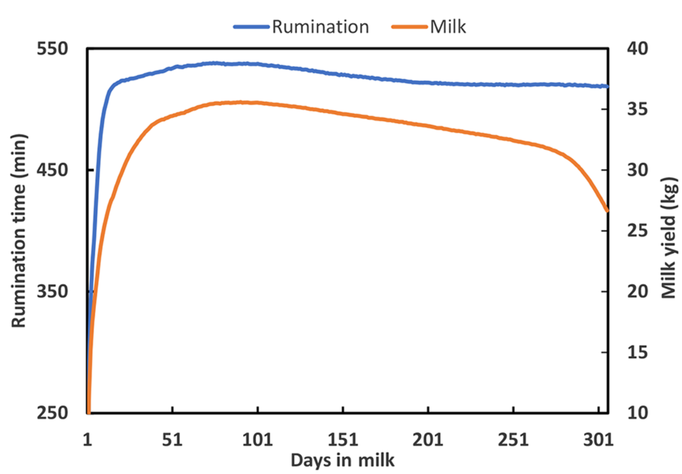

Figure 1. Mean daily rumination time and daily milk yield by DIM.

Table 3. Effects included in the analysis models and number of levels, records, and animals[1]

Analysis[2]	Traits analyzed	Number of cows with records	Effects included in the analysis models	Number of levels
2–4	5 disease traits and BCS	40,860	Additive genetic	84,659[3]
			Herd-year-season	1,407
			Calving ease	4
			Calving age	9
			Calving month	12
5–7	7 index traits	40,463	Additive genetic	83,435
			Herd-year-season	1,399
			Calving ease	4

[1]All analyses were computed by the individual animal model using the AIREMLf90 program (Misztal et al., 2014).

[2]The analyses also included the resilience indicator traits, as described in the text. Analysis 1, which included the means for both RT and milk, –LnVar, and autocorrelation for both traits, in both the first 40 DIM and the remainder of the lactation had the same effects and number of levels as analyses 2 to 4.

[3]Parents and grandparents of cows with records were included in the relation matrix.

Statistical Analysis

To compute partial lactation records for RT and milk, accounting for DIM, the individual daily records were analyzed PROC GLM of SAS (SAS Institute Inc., 2013) using the following linear fixed model:

$$T_{ij} = S_i + L_i + Q_i + N_i + e_{ij}, \quad [1]$$

where T_{ij} is the daily RT or milk record of cow j with i DIM; S_i, L_i, Q_i, and N_i are the square root, linear, quadradic, and inverse effects of i DIM; and e_{ij} is the random residual. The traits analyzed were the residuals of daily milk and RT derived from the analysis of Equation [1]. Previous studies found this model to better fit the data for partial lactation records than polynomials or class effects (Weller and Ezra, 2015, 2024).

As noted previously, records from 1 to 40 DIM and 41 to 305 DIM were analyzed separately. For each partial lactation record, milk production and RT from DIM 1 to 40 and from DIM 41 to 305, the natural logarithm of the variance of the residuals (**LnVar**) and their first order autocorrelations was computed (Keßler et al., 2025). Following Keßler et al. (2025) the negative of LnVar (−LnVar) was analyzed, as low variance should be associated with resilience.

All variance component analyses were based on the multitrait individual animal model. Given the large number of traits included in the analysis, it was necessary to analyze different groups of traits separately. Seven analyses were computed, and these are described in Table 3. The first analysis included 12 traits: the means, −LnVar, and autocorrelation for both RT and milk, in both the first 40 DIM and the remainder of the lactation. The second analysis included −LnVar and autocorrelation for both RT and milk in the first part of the lactation, the 5 disease traits, and BCS. The third analysis included −LnVar and autocorrelation for both traits in the later part of the lactation, the 5 disease traits, and BCS. The fourth analysis included the means of RT and milk in both the early and late parts of the lactation, the 5 disease traits, and BCS. Thus, analyses 2 to 4 each included 10 traits. Analyses 5, 6 and 7 included the same series of traits for RT and milk, but with the 7 selection index traits, instead of the disease traits and BCS. Thus, analyses 5 to 7 each included 11 traits. All models included a fixed herd-year-season effect and a fixed calving ease effect.

In addition to the additive genetic effect of animals with records, the analyses included all known parents and grandparents of cows with records. Two groups were defined for animals with unknown parents, one for males and one for females. Two seasons were defined for each herd-year, based on calving date: summer calvings from April 1 through September 30, and winter calvings beginning in the remaining months. Analyses 1 to 4 also included calving age in mo and calving mo as fixed effects. These effects were not included in analyses 5 to 7 because the index traits were preadjusted for these factors. In analyses 5 to 7, the RT and milk traits were preadjusted for these factors by linear model including these 2 fixed effects. The analyses were then performed on the residuals from the fixed model analyses. The effects included in the analysis models, and number of levels, records and animals are given in Table 3. The effects included and the number of levels in analysis 1 were the same as for analyses 2 to 4.

Variance components for all 7 analyses were first estimated using the MTC program (Misztal, 1994) to obtain preliminary estimates for the variance components. This program does not compute standard errors of the estimates, requires equal numbers of records for all traits, and assumes the same fixed effects for all traits. Therefore, for analysis of the 7 selection index traits, cow records were retained only if there were valid lactation records for milk, RT, and all 9 index traits. This resulted in 33,178 cow records. Most of the reduction was due to the lack of fertility records for cows that were not inseminated, and for persistency records of younger cows.

The data were then reanalyzed by the AIREMLf90 program (Misztal et al., 2014) using the variance component matrices from the MTC analyses as preliminary estimates. Standard errors were estimated for the variance components, the heritability estimates, and the estimates of the genetic and environmental correlations. For analyses 2 to 4, the datasets for AIREMLf90 analyses were the same as for MTC. Analyses 5 to 7 included 40,463 cows with records for the RT and milk resilience functions and valid records for the 3 milk production traits. This resulted in some missing values for the other index traits and the breeding index, and these records were denoted as missing in the analyses (in the Israeli milk recording system, all cows with valid records for milk also have valid records for fat and protein production).

The AIREMLf90 program also computes solutions for all effects, including the additive animal effects, which can be considered as breeding values for each trait. The breeding values of BCS, functions of RT and milk in DIM 1 to 40 were then used to predict the breeding values of the cows with records and sires with at least 10 daughters with records for the 5 disease traits, the 7 index traits, and PD20. The multiple regression model was as follows:

$$EBV_i = a \times BCS_i + b \times MM_i + c \times MR_i + d \times VM_i + f \times VR_i + g \times AM_i + h \times AR_i + e_i, \quad [2]$$

where EBV_i is the breeding value of cow i for the dependent variable (the breeding index, the index traits, and the 5 disease traits); BCS_i is the genetic evaluation for BCS of cow_i; MM_i is the genetic evaluation for the mean of milk production for DIM 1 to 40; MR_i is the genetic evaluation for the mean of RT for DIM 1 to 40; VM_i is the genetic evaluation for LnVar of milk; VR_i is the genetic evaluation for LnVar of RT; AM_i is the genetic evaluation for the autocorrelation of milk; AR_i is the breeding value for the autocorrelation for RT; a, b, c, d, f, g, and h are regression coefficients; and e_i is the random residual for cow i. Coefficients of determination were computed for all dependent variables. The analyses were also performed with BCS_i and the evaluations of the milk functions as independent variables, but without the RT functions. Thus, we were able to determine the added explanatory power of the 3 RT functions. The same models were also used to predict the breeding values of the sires, with the sire breeding values replacing the cow evaluations in all variables. For the 5 disease traits and BCS, the assumed breeding values were the solutions from models 2 and 5 for both cows and sires. For the index traits, the breeding values from the official October 2025 evaluation were used. The methods used to compute the breeding values are described by Weller and Ezra (2004, 2015, 2016) and Weller et al. (2006).

Table 4. Means and variances of functions of daily rumination time (RT, min) and daily milk production (kg) from 40,860 cow records[1]

DIM	Function	Trait	Mean	SD
1–40	Means[2]	RT	500.8	55.7
		Milk	29.2	5.0
	−LnVar	RT	−8.40	0.54
		Milk	−2.16	0.85
	Autocorrelation	RT	0.28	0.23
		Milk	0.31	0.28
41–305	Means	RT	528.1	44.0
		Milk	34.6	4.7
	−LnVar	RT	−8.27	0.37
		Milk	−2.50	0.62
	Autocorrelation	RT	0.28	0.17
		Milk	0.51	0.20

[1]−LnVar, the negative of the natural log of the daily variance, and the autocorrelations were computed from the residuals of Equation [1].

[2]Means of RT and milk are the phenotypic means of these traits, not the means of the residuals from Equation [1], which are arbitrary.

RESULTS

The means and variances of functions of daily RT and milk production are given in Table 4. The means of RT and milk derived from the residuals of Equation [1] are arbitrary. Therefore, the phenotypic means of these traits are presented. Means of autocorrelation were all positive. Standard deviations among cows were higher for the first part of the lactation for all traits. Means of −LnVar for milk were close to double the values of Poppe et al. (2021), but means of the autocorrelations were similar. However, values in Poppe et al. (2021) were computed for the complete lactation, and their means of daily milk production were 25% lower.

Table 5 shows the heritabilities and genetic and environmental correlations for means, −LnVar, and autocorrelations for daily RT and daily milk production for the first 40 DIM and the remainder of the lactation. Standard errors for heritabilities were all <0.016. Heritabilities for RT and milk are similar to previous results for this population (Weller and Ezra, 2024). Genetic correlations between means and −LnVar between the first and second parts of the lactation were >0.6 for both traits, and 0.4 and 0.6 for the autocorrelations. Heritabilities were generally higher for the second part of the lactation, which was >6 times longer than the first part. Heritabilities were higher for −LnVar than for the corresponding autocorrelations for both traits for both time periods. A positive correlation is generally obtained between the means and variances, and this was the case for milk (that is, a negative correlation between milk and −LnVar) but not RT. Absolute values of the genetic

Table 5. Heritabilities (on the diagonal in bold) and genetic (above the diagonal) and environmental (below the diagonal) correlations, for means, negative natural log of the variance (−LnVar), and autocorrelations (Autocorr) for rumination time (RT) and daily milk production (milk); based on data from 40,860 first-parity cows

			Days 1–40						Days 41–305					
			Means		−LnVar		Autocorr		Means		−LnVar		Autocorr	
DIM	Function	Trait	RT	Milk	RT	Milk	RT	Milk	RT	Milk	RT	Milk	RT	Milk
1–40	Means	RT	**0.338**	0.244	0.022	−0.050	−0.110	0.069	0.858	0.192	0.084	−0.152	−0.046	0.098
		milk	0.282	**0.333**	−0.001	−0.773	0.067	0.489	0.203	0.806	0.007	−0.790	−0.034	0.381
	LnVar	RT	0.280	0.301	**0.054**	0.026	−0.569	−0.195	−0.014	−0.008	0.646	0.085	−0.255	−0.118
		milk	0.057	0.102	0.176	**0.115**	−0.228	−0.426	−0.017	−0.714	0.072	0.809	−0.150	−0.263
	Autocorr	RT	−0.199	−0.182	−0.555	−0.091	**0.020**	0.325	0.027	0.196	−0.112	−0.214	0.421	0.213
		milk	−0.094	−0.185	−0.168	−0.156	0.146	**0.042**	0.126	0.573	−0.131	−0.585	0.103	0.600
41–305	Means	RT	0.480	0.160	−0.017	−0.040	−0.028	−0.018	**0.448**	0.212	0.132	−0.170	−0.055	0.133
		milk	0.127	0.614	0.116	0.000	−0.087	−0.064	0.243	**0.529**	0.067	−0.871	−0.039	0.302
	LnVar	RT	0.061	0.012	0.084	0.005	−0.037	−0.010	0.257	0.111	**0.168**	0.067	−0.610	−0.127
		Milk	−0.042	−0.177	0.000	0.099	0.009	−0.005	−0.016	−0.012	0.264	**0.179**	−0.040	−0.547
	Autocorr	RT	−0.049	0.012	−0.038	0.004	0.040	0.014	−0.208	−0.065	−0.676	−0.205	**0.119**	0.111
		Milk	0.002	−0.003	−0.024	−0.003	0.011	0.042	−0.049	−0.153	−0.197	−0.554	0.192	**0.049**

correlations between −LnVar and the corresponding autocorrelation were >0.5 for RT in both parts of the lactation, but slightly lower for milk.

Heritabilities and standard errors of the disease traits, BCS, and the index traits analyzed are given in Table 6. Heritabilities are similar to previous results for this population (Weller et al., 2019; Weller and Ezra, 2024, 2025). The exception was herd life, which in this analysis is based only on relatively young cows. As noted previously, for most records, actual herd life is unknown, and the analysis was based on the expectations. Heritabilities for retained placenta, displaced abomasum, and milk fever were <0.01.

Genetic correlations between disease traits, BCS and daily means, −LnVar, and autocorrelation of rumination and milk production, computed for 1 to 40 DIM and 41 to 305 DIM, are given in Table 7. Similarly, the genetic correlations between the selection index traits and daily means, −LnVar, and autocorrelation of rumination and milk production, computed for 1 to 40 DIM and 41 to 305 DIM, are given in Table 8. Values significantly different from zero ($P < 0.05$) are indicated. With respect to the disease traits, negative values indicate lower incidence of disease. Because the correlations are given with −LnVar, a negative value is indicative of higher resilience, as was the case for all the significant correlations.

Both positive and negative values were found for the autocorrelations. Significant genetic correlations with the 2 resilience indicator traits in the first part of the lactation were found for metritis, ketosis, and milk fever. For the later part of the lactation, none of the disease traits had significant correlations with the autocorrelation of either trait. Only one significant correlation was found for displaced abomasum with −LnVar in the later part of the lactation, but the heritability of this trait was <0.01. Among the disease traits the correlation with the highest absolute value was for −LnVar of milk in the first part of the lactation with ketosis, −0.62. That is, higher ketosis is associated with higher variance. Body condition score had significant correlations with all 3 milk functions in the early part of the lactation.

Positive values were economically favorable for all index traits except SCS. Significant correlations were found with all 3 functions of milk in the first part of the lactation for all traits except persistency. None of the index traits had significant genetic correlations with −LnVar or the autocorrelation of RT in either part of the lactation. The 3 milk production traits had negative genetic correlations with −LnVar of milk in the first part of the lactation. That is, higher variance was correlated with greater production. Both −LnVar and the autocorrelation of milk were correlated with female fertility. Higher mean RT in the first 40 DIM was positively correlated with milk, protein, and herd life.

Table 6. Heritabilities and their standard errors for the disease traits, BCS, and the index traits analyzed

Trait type	Trait	Heritability	SE
Disease	Retained placenta	0.006	0.002
	Metritis	0.059	0.006
	Ketosis	0.051	0.006
	Displaced abomasum	0.007	0.002
	Milk fever	0.002	0.001
	BCS	0.257	0.016
Index	Milk	0.354	0.027
	Fat	0.268	0.027
	Protein	0.259	0.023
	SCS	0.160	0.016
	Female fertility	0.044	0.007
	Herd life	0.051	0.008
	Persistency	0.138	0.015

Table 7. Genetic correlations between disease traits and BCS with the daily means, the negative natural log of the variance (–LnVar), and autocorrelation (autocorr) of rumination time (RT) and milk production (milk), computed for 1–40 DIM and 41–305 DIM (40,860 first-parity cows)

	1–40 DIM						41–305 DIM					
	Means		−LnVar		Autocorr		Means		−LnVar		Autocorr	
Trait	RT	Milk	RT	Milk	RT	Milk	RT	Milk	RT	Milk	RT	Milk
Retained placenta	0.048	0.542*	0.095	−0.290	0.121	0.385	0.102	0.552*	−0.009	−0.395*	0.227	0.221
Metritis	−0.165	0.063	−0.204*	−0.312*	0.395*	0.327*	−0.076	0.156*	−0.058	−0.211	0.093	0.105
Ketosis	−0.345*	0.389*	−0.097	−0.623*	0.269*	0.355*	−0.179*	0.431*	−0.046	−0.414*	0.099	0.011
Displaced abomasum	−0.103	−0.008	−0.218	−0.247	0.100	0.352	−0.033	0.130	−0.033	−0.362*	−0.165	0.193
Milk fever	−0.299*	−0.176	−0.151	−0.353*	0.165	−0.447*	−0.487*	−0.091	−0.325*	−0.292	−0.038	−0.092
BCS	−0.164*	−0.151*	−0.068	0.350*	−0.059	−0.288*	−0.121*	−0.043	−0.073	0.237*	0.002	−0.172*

*$P < 0.05$.

Coefficients of determination and the effect with the greatest t-value for the breeding values of the disease and index traits as computed from the models of Equation [2] are given in Table 9 for cows with breeding values and sires with at least 10 daughters with valid records. Because only coefficients of determination and t-values are presented, the sign of LnVar is immaterial. Coefficients of determination are presented first for the model that included BCS and the functions of daily milk, and then for the model that included BCS and the functions of both daily milk and RT. For all traits, inclusion of the RT traits increased the coefficient of determination.

For the cow evaluations, nearly all of the coefficients for the resilience trait evaluations were significantly different from zero at $P < 0.001$ for all the disease and index traits. Body condition score near calving was significant for all disease and index traits ($P < 0.001$). The coefficients of determination for the disease traits were all >0.35, and >0.6 for 3 diseases. The coefficient of determination was highest for ketosis, which also had the highest correlation with the resilience traits in Table 7. For all the disease traits, the resilience indicators with the highest t-value was either −LnVar or an autocorrelation, but not the trait means. Among the index traits, coefficient of determination was highest for milk, 0.54; but equal to 0.18 for the complete index. Milk has a coefficient of zero in the index, but is highly correlated with protein production. Coefficients of determination were lowest for SCS, herd life, and persistency.

Because the analyses of the sire evaluations were based on only 526 animals, as compared with 40,860 cows for the disease traits and slightly fewer for the index traits, most of the factors included in the sire models were not statistically significant. Coefficients of determination for the sire evaluations were lower for most traits, but higher for SCS, herd life, and the 2 calving traits. Although coefficients of determination of the sire evaluations were lower for all disease traits, all values were >0.25. As was the case for the cow evaluations, inclusion of the RT functions increased the coefficient of determination for all traits analyzed. Of the 14 traits analyzed, the same factor had the largest t-value for both the cow and sire evaluations for 8 traits. The regression coefficients for −LnVar for the selection index for both cows and sires were in the desired direction. That is, lower variance, greater resilience and higher breeding values; however, for milk yield, the coefficients for −LnVar were negative, in accordance with the values in Table 8. The coefficient of determination for sires the breeding index was 0.19 as compared with 0.18 for cows.

DISCUSSION

The current study is unique in that in addition to daily milk yield, functions of daily RT were also analyzed. In addition, mean daily milk production and RT were included as predictors of the disease and index traits. Because these traits do not measure variability in the traits over time, they can be considered negative controls. For retained placenta, only mean milk production had a significant correlation, but the heritability of this trait is <1%. Significant correlations with −LnVar and the autocorrelation of milk were found for milk fever, even though this trait also has heritability <1%. In general, for the disease traits, −LnVar of milk in the first 40 DIM was a better indicator trait than either mean milk or RT. This was not the case for the index traits.

Keßler et al. (2025) found significant correlations between LnVar of daily milk production and metritis, ketosis, displaced abomasum, milk fever, milk yield, and herd life. The highest correlation was with ketosis, 0.36, as compared with 0.62 in the current study for the first 40 DIM. All of their autocorrelations were <0.15, and their study was based on 7,623 lactations from parities 1 to 6. They concluded that a resilience selection index composed of 2 different variance-based resilience indicator traits was optimal. Elgersma et al. (2018), in an analysis of 67,025 first-parity lactations, found the strongest cor-

Table 8. Genetic correlations between index traits and daily means, the negative natural log of the variance (−LnVar), and autocorrelation (autocorr) of rumination time (RT) and milk production, computed for 1–40 DIM and 41–305 DIM (40,463 first-parity cows)

	1–40 DIM						41–305 DIM					
	Means		−LnVar		Autocorr		Means		−LnVar		Autocorr	
Trait	RT	Milk	RT	Milk	RT	Milk	RT	Milk	RT	Milk	RT	Milk
Milk	0.217*	0.807*	−0.013	−0.660*	0.197	0.452*	0.306*	0.996*	0.111	−0.844*	0.010	0.295
Fat	0.148*	0.575*	0.112	−0.324*	−0.056	0.029	0.149	0.390*	0.055	−0.348*	−0.080	0.089
Protein	0.361*	0.744*	0.074	−0.514*	0.025	0.243*	0.398*	0.804*	0.070	−0.675*	−0.039	0.214
SCS	−0.033	0.178*	0.026	−0.222	−0.020	0.319*	−0.042	0.143*	0.017	−0.220*	0.149	0.248
Fertility	−0.054	−0.428*	0.110	0.377*	−0.128	−0.403*	−0.039	−0.379*	0.037	0.372*	0.049	−0.192
Herd life	0.248*	0.078	0.179	0.022	−0.169	−0.361*	0.321*	0.249*	0.155	−0.002	−0.092	−0.191
Persistency	0.037	−0.092	0.058	−0.040	0.069	−0.082	0.195*	0.261*	0.125	−0.049	−0.097	−0.280

*$P < 0.05$.

relations (−0.29 to −0.52) between LnVar of daily milk yield and udder health, ketosis, persistency, and longevity. They found a heritability of 0.1 for LnVar for the complete lactations as compared with 0.11 for DIM 1 to 40 and 0.18 for DIM 41 to 305 in the current study.

Poppe et al. (2021) divided 200,084 complete first-parity lactations into 3 periods of 100 d each. They found very similar heritabilities for LnVar of daily milk production for the 3 periods (0.12–0.15). Similarly, the heritabilities for the autocorrelations ranged from 0.05 to 0.06. For the complete first lactation, they found heritabilities of 0.20 and 0.08 for LnVar and the autocorrelation, which were somewhat higher than in the current study. Genetic correlations among the lactation periods were all >0.6 for both traits. In the current study, the genetic correlation between the 2 periods was 0.8 for −LnVar and 0.6 for the autocorrelation. It should be noted that in the current study, the first period was only 40 DIM, as compared with 100 DIM in Poppe et al. (2021).

Chen et al. (2024) found heritabilities of 0.13 and 0.04 for LnVar and autocorrelation of daily milk production, respectively, for complete first lactation records of 11,138 US Holsteins. As in the other studies cited, absolute values of the genetic correlations of the resilience indicators with economic traits were higher for LnVar than for the autocorrelation. First-parity genetic correlations of LnVar with milk production, productive herd life, and daughter pregnancy rate were 0.50, −0.30, and 0.32, respectively. The results for milk and fertility are slightly lower than the results in the current study for 1 to 40 DIM. Elgersma et al. (2018) found a heritability of 0.10 for first-parity LnVar based on 67,025 lactations of Dutch Holsteins. Genetics correlations of LnVar with udder health index, ketosis, longevity, and persistency were −0.36, −0.52, −0.30 and −0.29, respectively. Thus, the results for ketosis and persistency in the latter part of the lactation were similar to the results of the current study. Wang et al. (2022), in an analysis of 6,816 Chinese Holstein cows, found that log-transformed SD of milk deviations, when removing the first and last 10 d of the lactation, had a heritability of 0.25 and was associated with less milk losses, better reproductive performance, and lower disease incidence. A genetic correlation of 0.87 was obtained with "udder health."

Shook (1989) proposed that a potential trait must meet several criteria for inclusion in the selection objective. First, it should have an economic value. Second, the trait must have sufficiently large genetic variation and heritability. Third, the trait should be clearly defined, measurable at a low cost, and consistently recorded. An indicator trait may be favored if it has a high genetic correlation with an economically important trait, reduces recording costs, has a higher heritability, or can be measured earlier in life. Inclusion of a resilience indicator trait based on daily milk production has the advantage that this trait is now routinely recorded in Israel for all cows, whereas RT and BCS are not.

During the last 30 years, the focus of selection has moved away from increasing milk production and toward traits related to herd life, fertility, calving, health, and workability. Current research focuses on fitness, health, welfare, and milk quality, in addition to environmental sustainability (Miglior et al., 2017). Miglior et al. (2017) predicted that in the future, on-farm sensors, data loggers, precision measurement techniques, and other technological aids will provide even more data for use in selection, and the difficulty will lie not in measuring phenotypes, but rather in choosing which traits to include in selection indices. Selection indices of the advanced countries are often updated. For example, "early first calving" and "heifer livability" were added to the main US index (NM$) in 2021 (Guinan et al., 2023). Bengtsson et al. (2022) showed that many modern dairy cattle breeding goals most likely have negative genetic trends for promising resilience indicators such as LnVar. Adding breeding goal weight to resilience indicators, such as BCS and LnVar, could reverse the negative trend observed for resilience indicators.

Table 9. Coefficients of determination of the breeding values of the cows with records for the disease and index traits and sires with at least 10 daughters with records as functions of the means for milk and rumination time (RT), the natural log of the variance (LnVar), and autocorrelation (autocorr) of milk and of rumination time, computed from 1 to 40 DIM and BCS near calving[1]

		Cows				Sires (526 animals)		
			Coefficient of determination			Coefficient of determination		
Trait group	Trait	Number of cows	BCS, milk	BCS, milk, RT	Effect with greatest *t*-value	BCS, milk	BCS, milk, RT	Effect with greatest *t*-value
Disease	Retained placenta	40,860	0.346	0.482	LnVar, RT	0.260	0.353	LnVar, RT
	Metritis	40,860	0.371	0.424	Autocorr, RT	0.225	0.271	BCS
	Ketosis	40,860	0.659	0.676	LnVar, milk	0.386	0.423	LnVar, milk
	Displaced abomasum	40,860	0.457	0.602	Autocorr, milk	0.313	0.381	BCS
	Milk fever	40,860	0.574	0.659	Autocorr, milk	0.425	0.510	Autocorr, milk
Index	Milk	40,598	0.507	0.538	Autocorr, milk	0.435	0.463	Autocorr, milk
	Fat	40,598	0.173	0.178	Mean, milk	0.128	0.140	Mean, milk
	Protein	40,598	0.283	0.340	Mean, RT	0.258	0.311	Mean, RT
	SCS	40,604	0.046	0.061	BCS	0.084	0.094	Autocorr, milk
	Fertility	38,007	0.128	0.137	Autocorr, milk	0.107	0.115	Autocorr, milk
	Herd life	40,732	0.022	0.041	BCS	0.075	0.099	Mean, RT
	Persistency	39,321	0.043	0.057	Mean, milk	0.020	0.060	Mean, RT
	Calving ease	39,984	0.113	0.128	Mean, milk	0.165	0.217	Autocorr, RT
	Calf survival	39,984	0.123	0.153	Mean, milk	0.154	0.176	Mean, milk
	Selection index	**35,929**	**0.165**	**0.182**	**Mean, milk**	**0.152**	**0.187**	**Mean, milk**

[1]Number of cows with valid records and results for the selection index are listed in bold type.

Genomic evaluation based on individual genotypes is becoming routine for female calves, as well as male calves, and reliabilities for animals without records are now substantially higher than those obtained based only on pedigree, including traits with low heritabilities (Guinan et al., 2023; Curzon et al., 2024). Because all the metabolic disease traits have low heritability and most of the index traits require complete lactation records for useful breeding values, inclusion of a resilience predictor trait based on the first 40 DIM in the selection index should be justified to improve selection decisions for young cows and sires, in addition to the predicted gain in resilience.

CONCLUSIONS

Significant genetic correlations with −LnVar and the autocorrelation of daily milk production in the first part of the lactation were found for metritis, ketosis, and milk fever. Among the disease traits, the correlation with the highest absolute value was for −LnVar of milk in the first part of the lactation with ketosis (−0.62). For the regression of cow evaluations on the breeding values of the resilience indicator traits, nearly all of the resilience traits were significant at $P < 0.001$ for the disease and index traits. Coefficients of determination for the sire evaluations were all >0.25 for the disease traits. Inclusion of the RT functions increased the coefficient of determination for all traits analyzed for both cows and sires. Inclusion of a resilience predictor trait based on the first 40 DIM in the selection is justified to improve selection decisions for young cows and sires.

NOTES

This study received no external funding. We thank Ignacy Misztal and Shogo Tsuruta (University of Georgia, Athens, GA) for the use of the MTC and AIREMLF90 programs, and Boas Hanochi (Israeli Cattle Breeders Association, Caesarea Industrial Park, Israel) for data validation. No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: Autocorr = autocorrelation; LnVar = natural logarithm of the variance of the residuals; PD20 = Israeli breeding index; RT = rumination time.

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